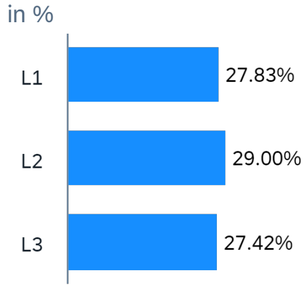
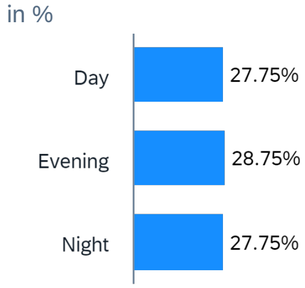
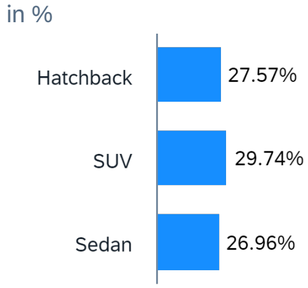
Observed Defect Rate by Production Line



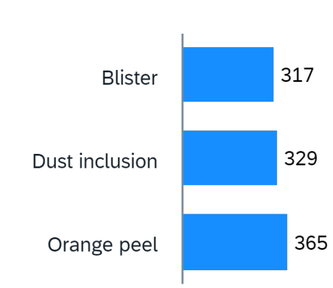
Observed Defect Rate by Shift



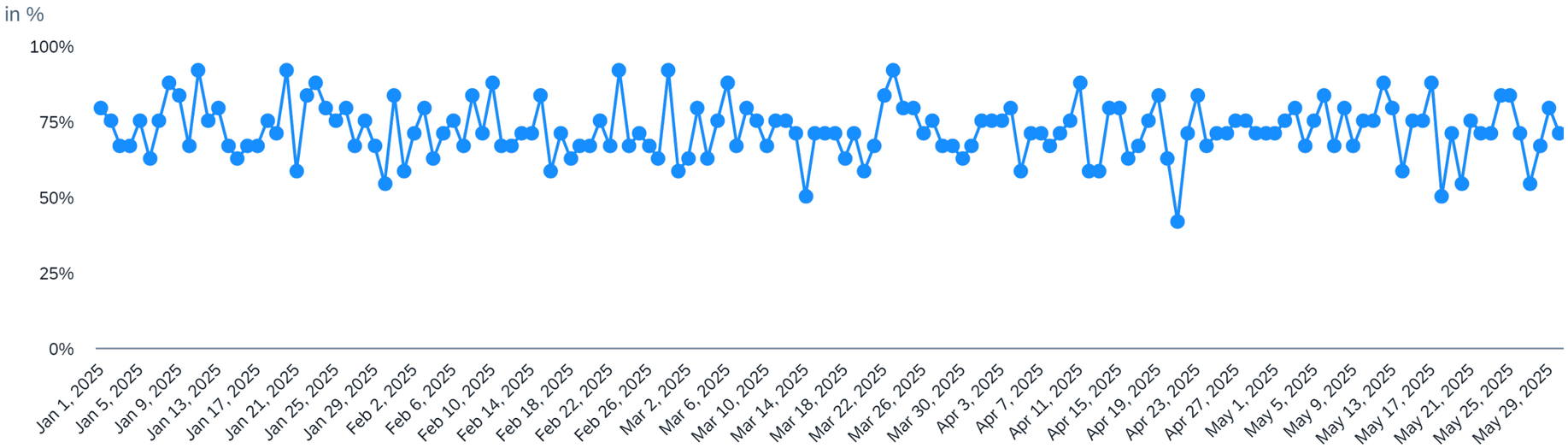
Observed Defect Rate by Vehicle Model



Failed Batches by Defect Category



First-Pass Yield Over Time



Estimated Batch Risk vs Actual Inspection

RISK_SCORE | 1

					Measures	RISK_SCORE	FAILED_BATCHES
BATCH_ID	**DEFECT_CATEGORY**	**LINE_ID**	**SHIFT_ID**	**TEMPERATURE_C**	**HUMIDITY_PCT**		
B00975	Orange peel	L1	Day	21.160794896667504	91.80562316824287	0.68452153	1
B02557	Orange peel	L2	Day	24.73516742473609	80.4699166826319	0.66710243	1
B02133	None	L1	Evening	24.165564684325	83.75177757563978	0.66139616	0
B01931	None	L3	Day	24.062221209327674	86.45288503697934	0.65805846	0
B02709	None	L1	Evening	23.10976745055683	82.26053401967987	0.65566119	0
B01156	Orange peel	L2	Night	25.58561988892308	79.32276584107873	0.65060862	1

Risk score: estimated defect risk (0–1), used to prioritise inspection—not proof of a defect. Actual inspection: 0 = passed; 1 = failed. Data is synthetic.

Downtime Analysis

